Suggested problems in §14.4: 1-45 odd, 49-53 odd, 57, 59, 69-73 odd

Problem 1: Compute the gradient of the function

$$f(x, y) = y \ln(x) + xy^2$$

at the point $(1, 2)$.

We have

$$\nabla f(x, y) = (y/x + y^2)\vec{i} + (\ln(x) + 2xy)\vec{j},$$

so

$$\nabla f(1, 2) = 6\vec{i} + 4\vec{j}.$$

Problem 2: Compute the gradient of the function

$$f(x, y) = \ln(x^2 + xy)$$

at the point $(4, 1)$.

We have

$$\nabla f(x, y) = \left(\frac{2x + y}{x^2 + xy}\right)\vec{i} + \left(\frac{x}{x^2 + xy}\right)\vec{j}$$

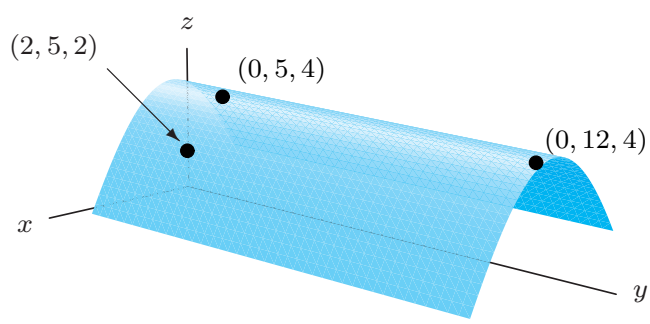
so

$$\nabla f(4, 1) = \left(\frac{9}{20}\right)\vec{i} + \left(\frac{1}{5}\right)\vec{j}.$$

Problem 3: The surface $z = f(x, y)$ is shown below. Suppose that

$$\vec{u} = \frac{1}{\sqrt{2}}(\vec{i} + \vec{j}) \quad \text{and} \quad \vec{v} = \frac{1}{\sqrt{2}}(\vec{i} - \vec{j})?$$

Are the following directional derivatives positive, negative, or zero?



(a) $f_{\vec{u}}(2, 5) < 0$

(b) $f_{-\vec{u}}(2, 5) > 0$

(c) $f_{\vec{v}}(2, 5) < 0$

(d) $f_{-\vec{v}}(2, 5) > 0$

(e) $f_{\vec{i}}(2, 5) < 0$

(f) $f_{\vec{j}}(2, 5) = 0$

(g) $f_{\vec{u}}(0, 5) < 0$

(h) $f_{-\vec{u}}(0, 5) < 0$

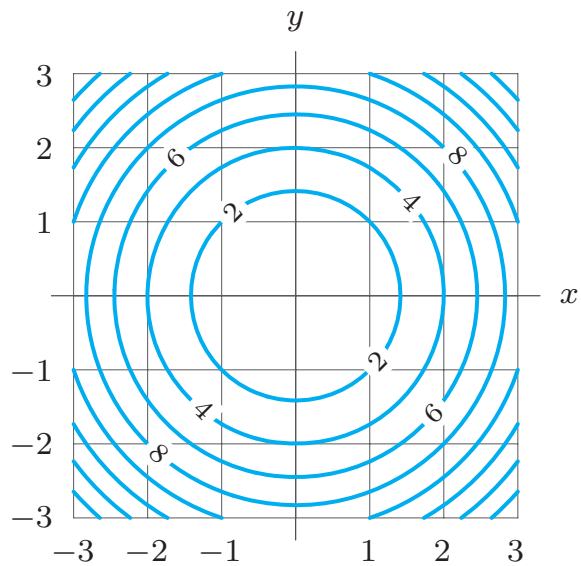
(i) $f_{\vec{v}}(0, 5) < 0$

(j) $f_{-\vec{v}}(0, 5) < 0$

(k) $f_{\vec{i}}(0, 5) < 0$

(l) $f_{\vec{j}}(0, 5) = 0$

Problem 4: Use the contour diagram of f shown below to determine whether the directional derivative at the indicated point in the indicated direction is positive, negative, or approximately zero.



(a) At $(-2, 2)$, in the direction $\vec{i} < 0$

(b) At $(0, -2)$, in the direction $\vec{j} < 0$

(c) At $(0, -2)$, in the direction $\vec{i} + 2\vec{j} < 0$

(d) At $(0, -2)$, in the direction $\vec{i} - 2\vec{j} > 0$

(e) At $(-1, 1)$, in the direction $\vec{i} + \vec{j} \approx 0$

(f) At $(-1, 1)$, in the direction $-\vec{i} + \vec{j} > 0$

Problem 5: If $f(x, y) = xy + y^3$ and $\vec{u} = \frac{1}{5}(3\vec{i} - 4\vec{j})$, find $f_{\vec{u}}(1, 2)$.

We have

$$f_{\vec{u}}(x, y) = \nabla f(x, y) \cdot \vec{u} = \frac{1}{5}(y\vec{i} + (x + 3y^2)\vec{j}) \cdot (3\vec{i} - 4\vec{j}) = \frac{1}{5}\left(3y - 4x - 12y^2\right).$$

Hence

$$f_{\vec{u}}(1, 2) = -\frac{46}{5}.$$

Problem 6: If $f(x, y) = 3x^2 - 2y^2$, find the instantaneous rate of change of f at $(-3/4, 0)$ in the direction from $(-3/4, 0)$ to $(0, 1)$.

First get a vector in the correct direction:

$$(3/4)\vec{i} + \vec{j}.$$

Now normalize it to get the unit vector

$$\vec{u} = \frac{4}{5}\left((3/4)\vec{i} + \vec{j}\right).$$

Then

$$f_{\vec{u}}(x, y) = \nabla f(x, y) \cdot \vec{u} = \frac{4}{5}(6x\vec{i} - 4y\vec{j}) \cdot ((3/4)\vec{i} + \vec{j}) = \frac{4}{5}\left(\frac{9}{2}x - 4y\right).$$

Hence

$$f_{\vec{u}}(-3/4, 0) = -\frac{27}{10}.$$